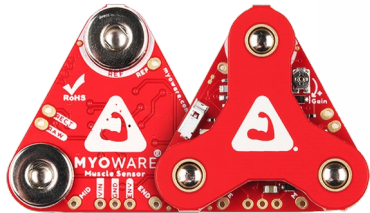
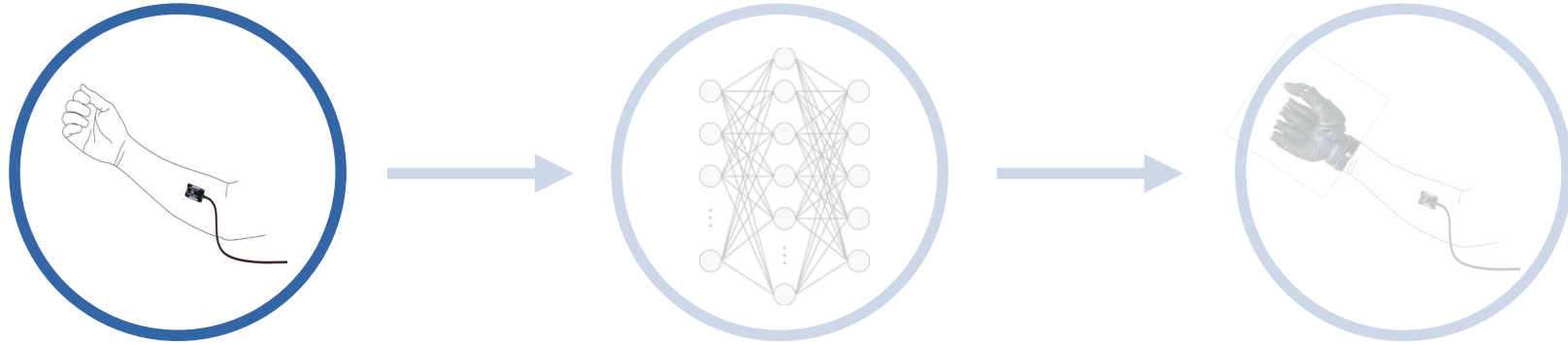


AI-Based Control System for Hand Gesture Prediction of RYO Smart Prosthetic Hand

Barret Feusi

Primary Objective



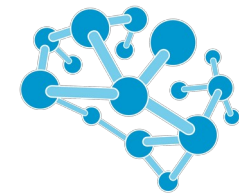
EMG



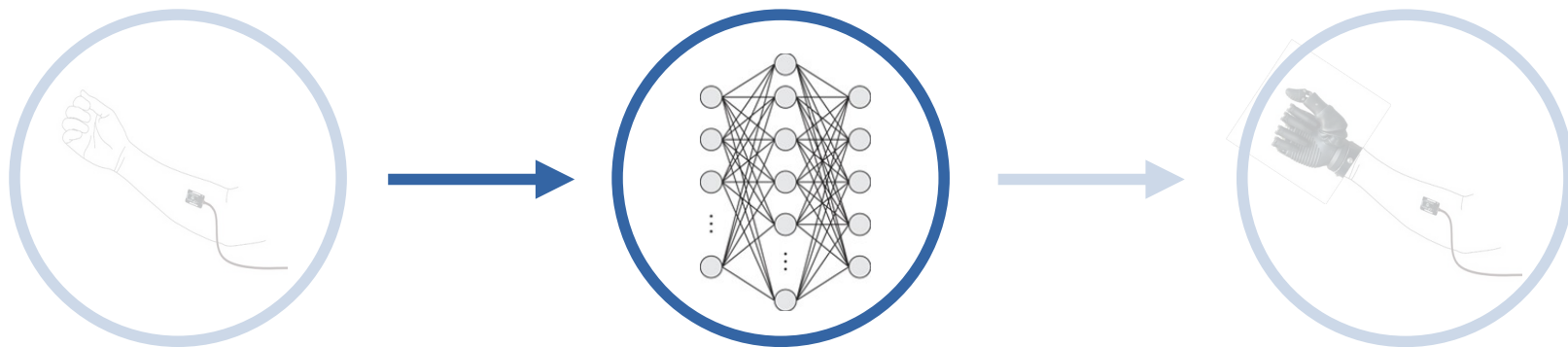
ESP32-S3

To make a “real-time [biosignal] acquisition and processing module...”

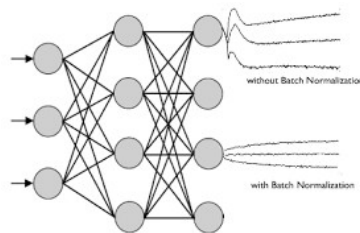
Primary Objective



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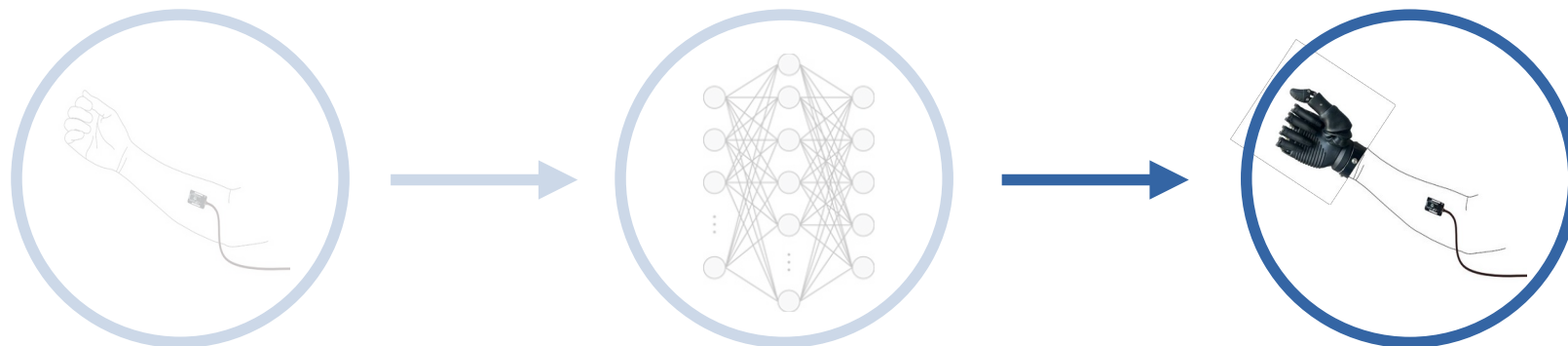


...with robust filtering

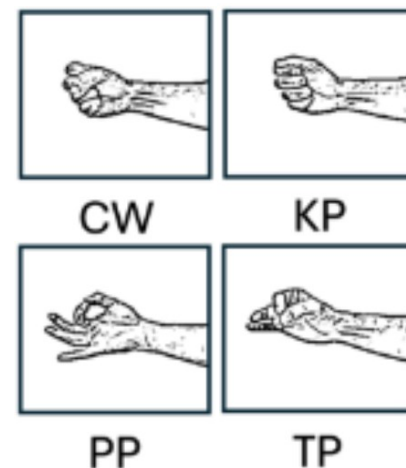


and normalization”...

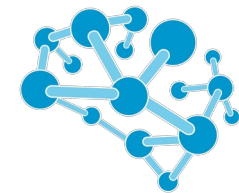
Primary Objective



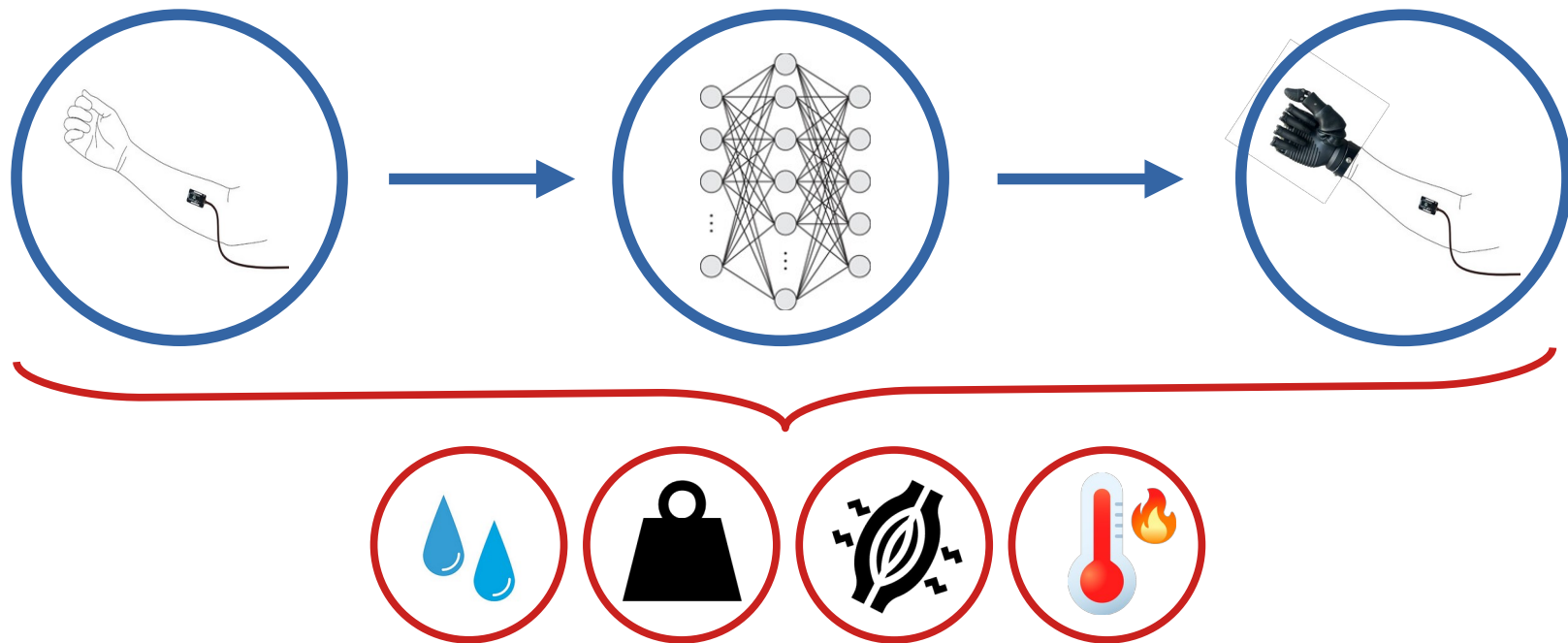
...to predict one of four hand gestures
for controlling an **affordable**
smart hand prosthesis.



Primary Objective



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AND have it work under non-ideal conditions!

Justification

“Today, daily life is composed of many computing systems, therefore interacting with them in a natural way makes the communication process more comfortable.... One form of [Human-Computer Interface] is Hand Gesture Recognition (HGR), which predicts the class and the instant of execution of a given movement of the hand. One possible input for these models is surface electromyography (EMG), which records the electrical activity of skeletal muscles. EMG signals contain information about the intention of movement generated by the human brain.”¹

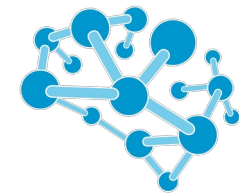
Justification

Hand gesture classification is crucial for proper control of human-assistive robotics, such as smart prostheses.²

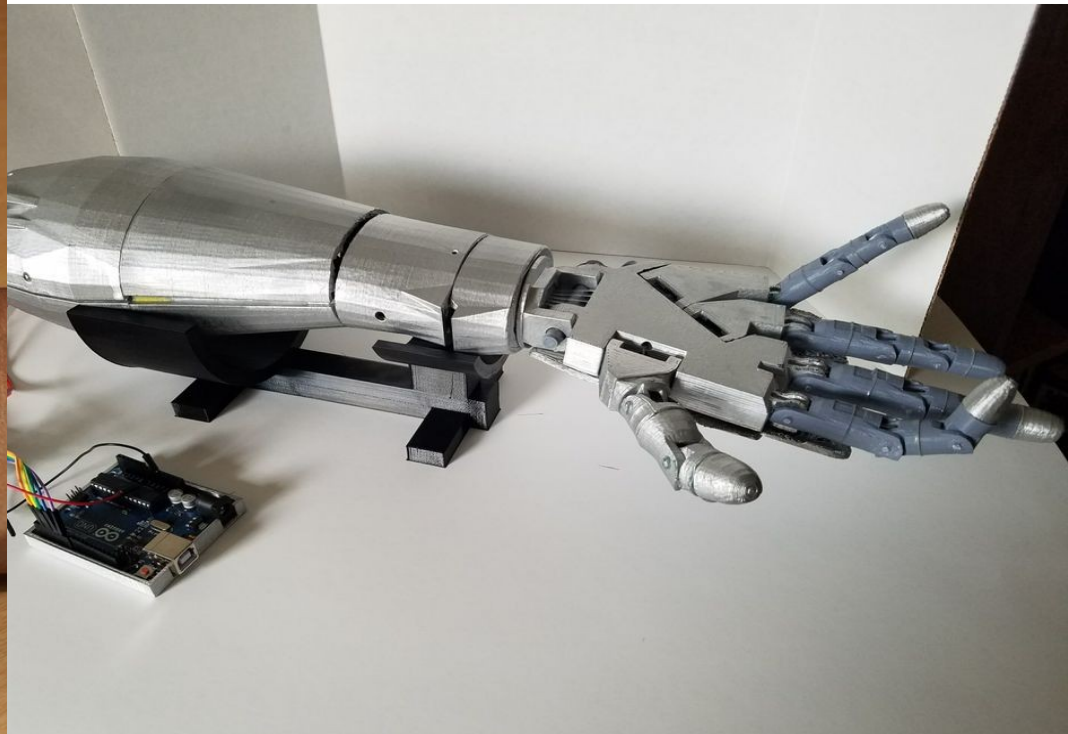
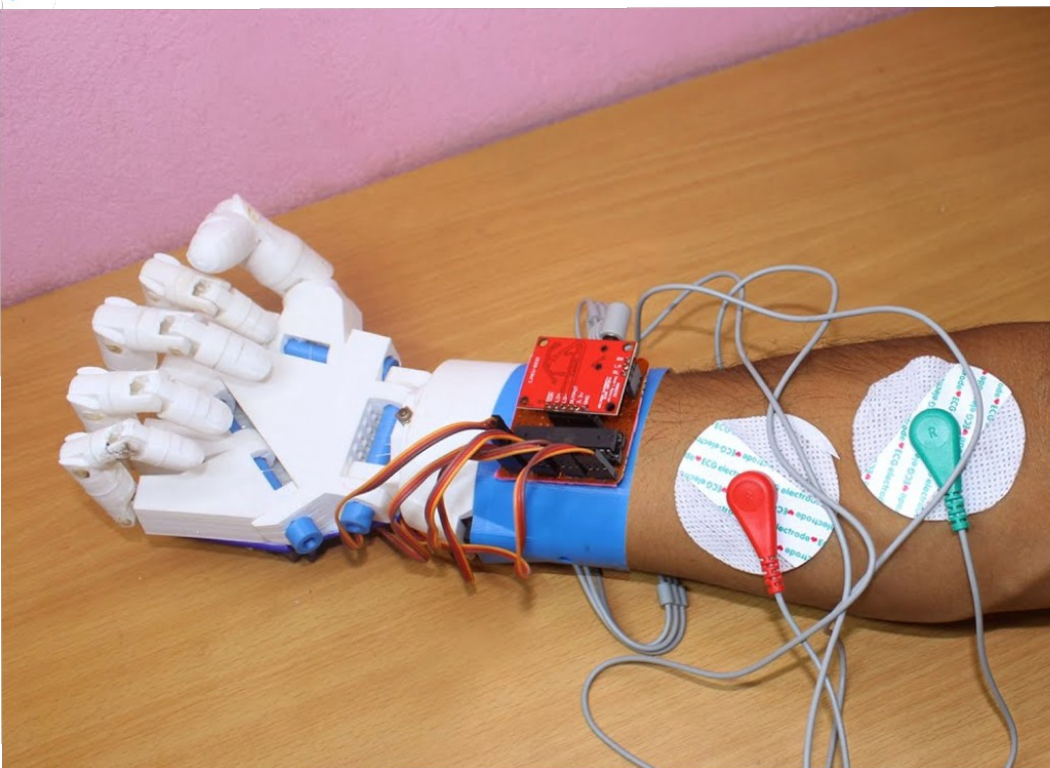
Ownership of these devices starts with efficient control.

Development Plan

Smart Prosthetic Arm

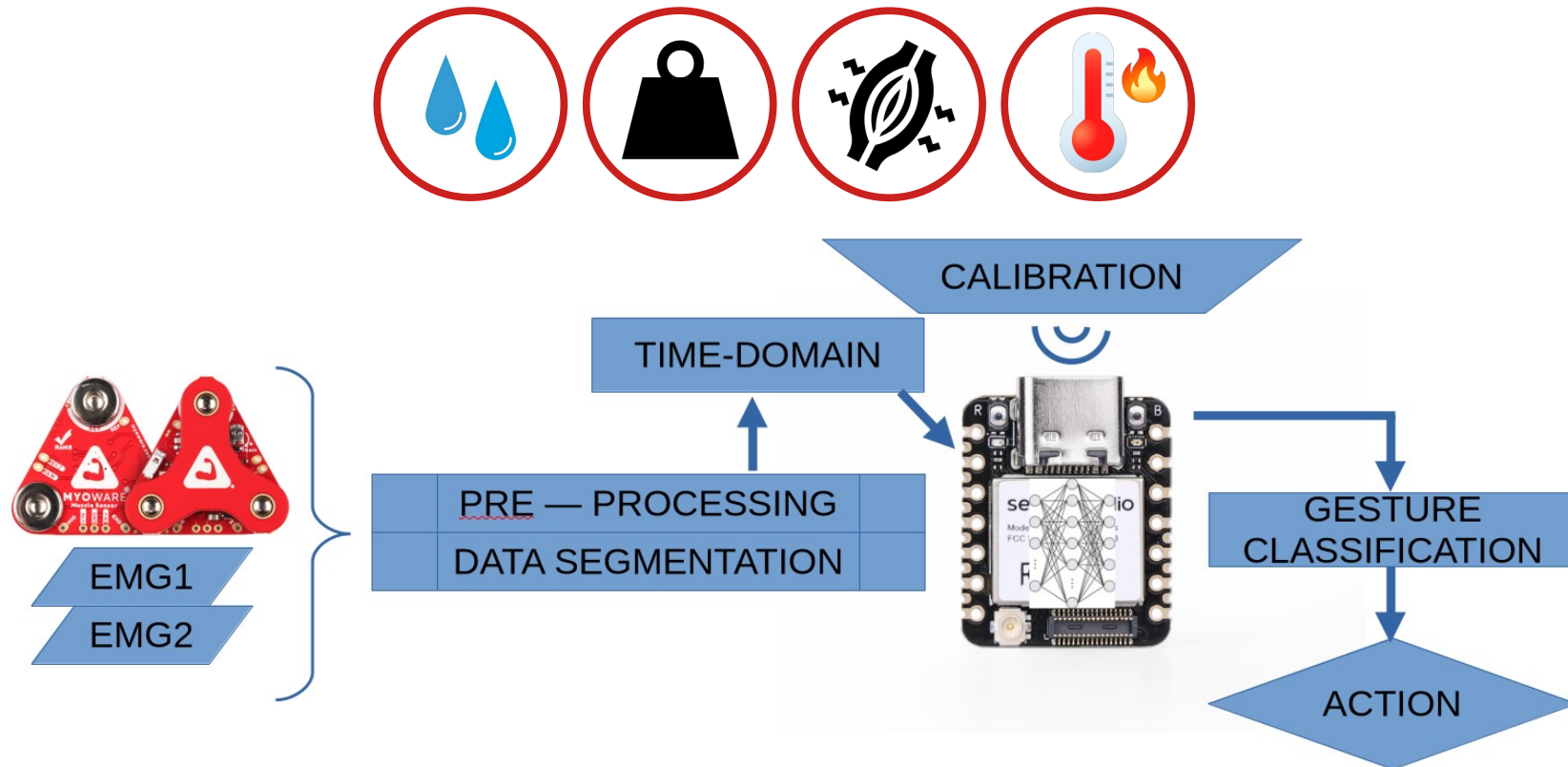


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Development Plan

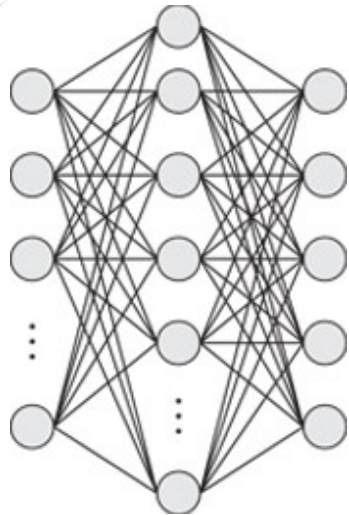
Electronics and Data Processing



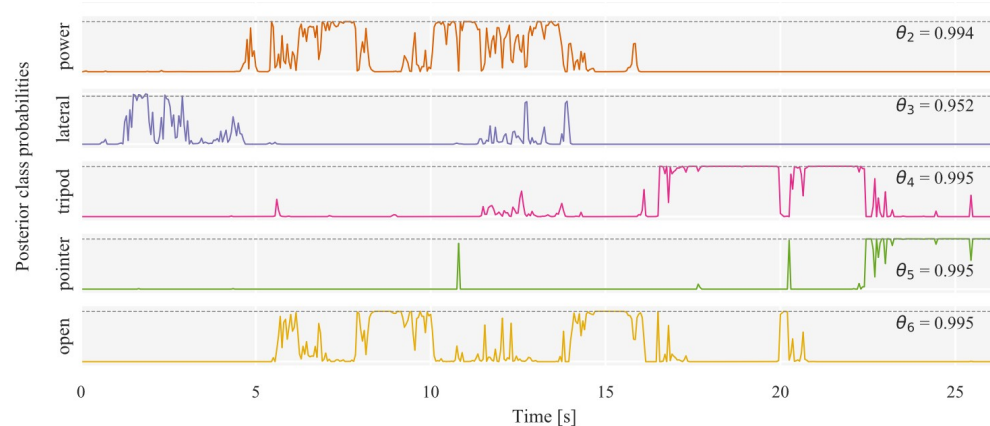
Development Plan



5 1-v-All RDA + 1D Dilated CNN ML Model

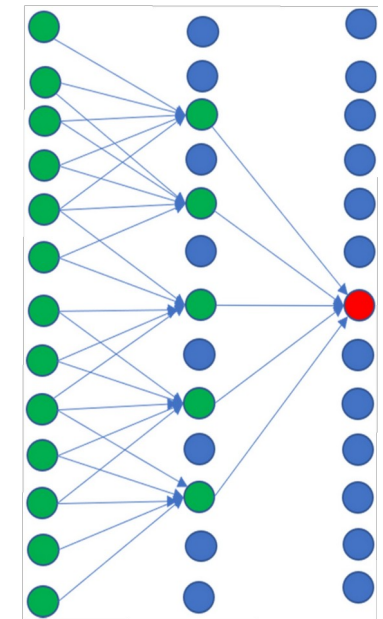


=



“Gatekeeper” Model

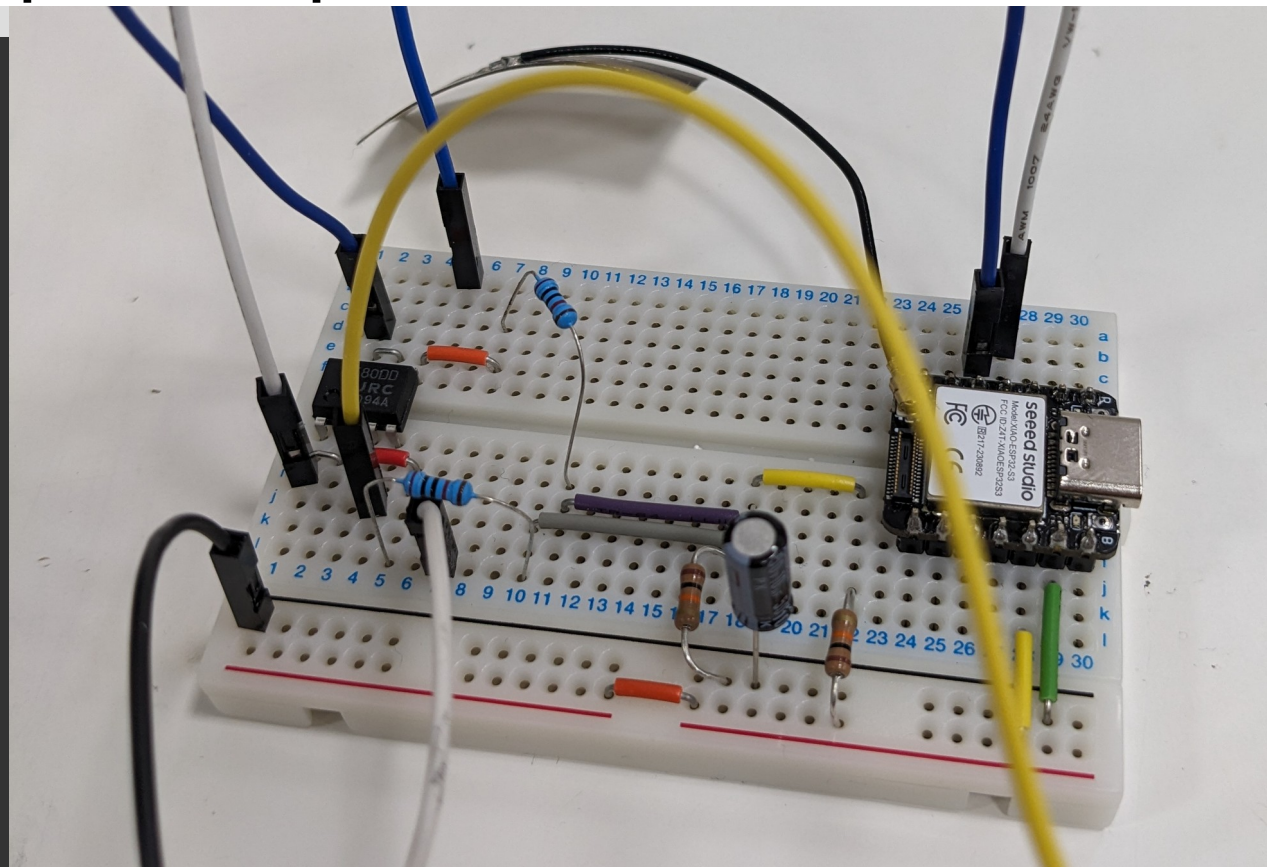
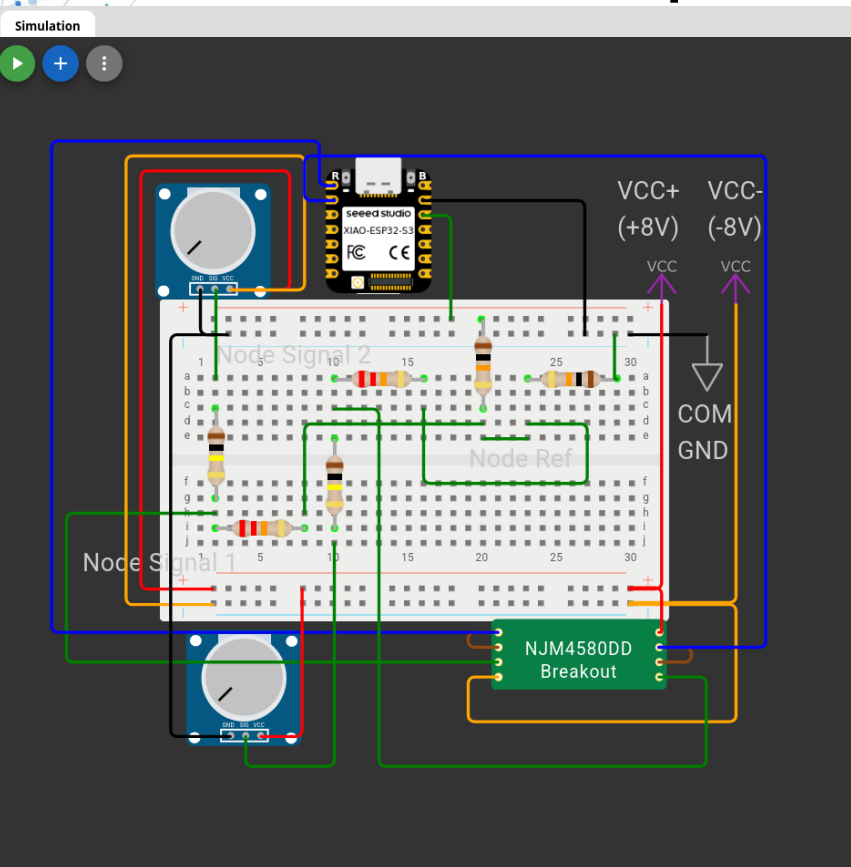
+



Second Opinion Model

Progress to Date

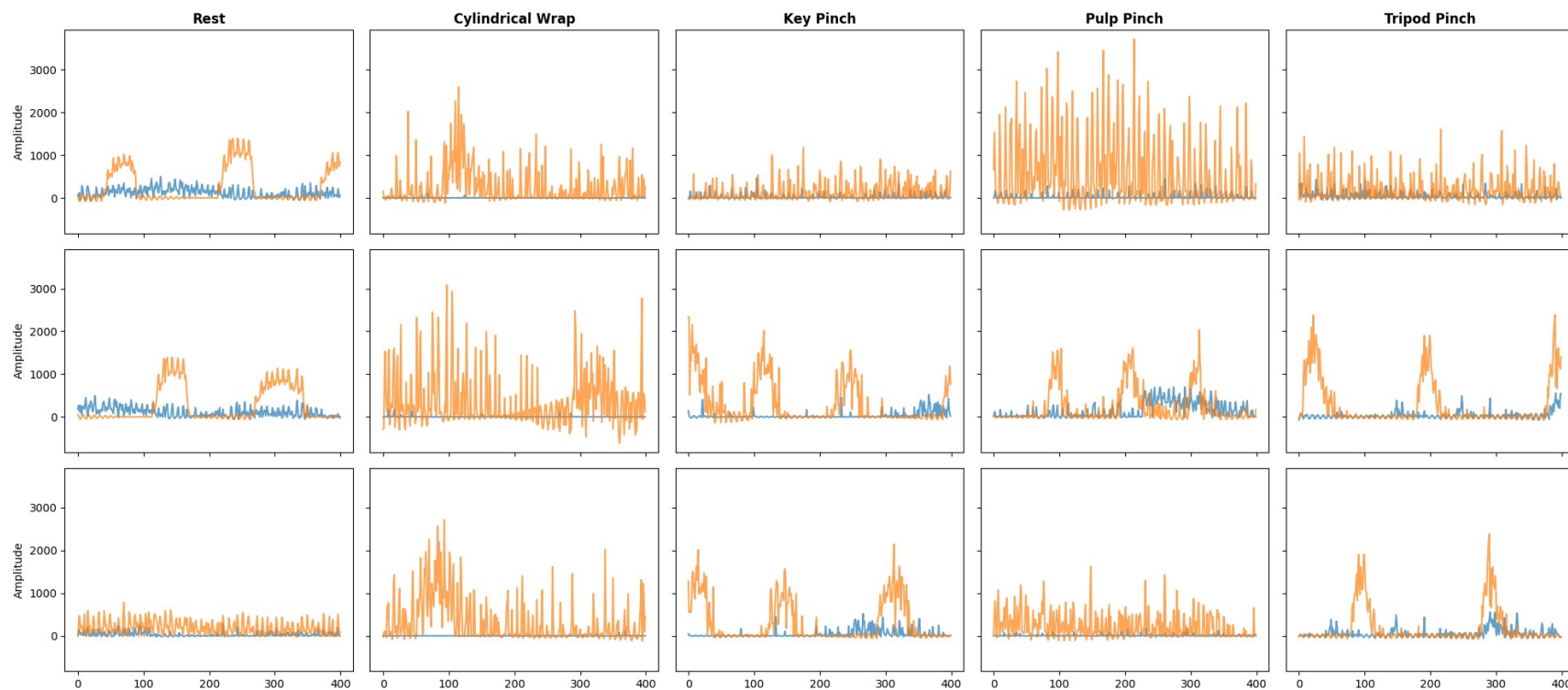
Developed Amplifier Setup for EMG Sensors



Progress to Date

Took Preliminary EMG Samples

Filtered sEMG Windows (60Hz Notch Applied)



Progress to Date



Created 5 1-v-All RDA “Gatekeeper” Models

```
Tuning RDA shrinkage parameter...
Best shrinkage parameter selected: 0.0
Calculating confidence-based rejection thresholds...
Acceptance Rate: 0.2036
Accuracy on Accepted Predictions: 1.0000
Used 57 out of 280 total possible segments to be predicted.
Class 0 Threshold: 0.9193
Class 1 Threshold: 0.6330
Class 2 Threshold: 0.7508
Class 3 Threshold: 0.7473
Class 4 Threshold: 0.7169

Average Conditional Accuracy across 10 folds: 1.0000

--- Final Model Showcase (10 Random Samples from Fold 10 Validation Set) ---
Actual | Predicted | Prediction | Confidence | Status
-----|-----|-----|-----|-----
Rest | N/A | Rest | 0.7967 | N/A
Rest | N/A | Key | 0.7234 | N/A
Rest | N/A | Rest | 0.8593 | N/A
Rest | N/A | Tripod | 0.4465 | N/A
Key | N/A | Key | 0.6255 | N/A
Rest | N/A | Rest | 0.8428 | N/A
Pinch | N/A | Rest | 0.4452 | N/A
Rest | N/A | Rest | 0.5885 | N/A
Key | Key | Key | 0.7545 | Yes
Rest | Rest | Rest | 0.9640 | Yes

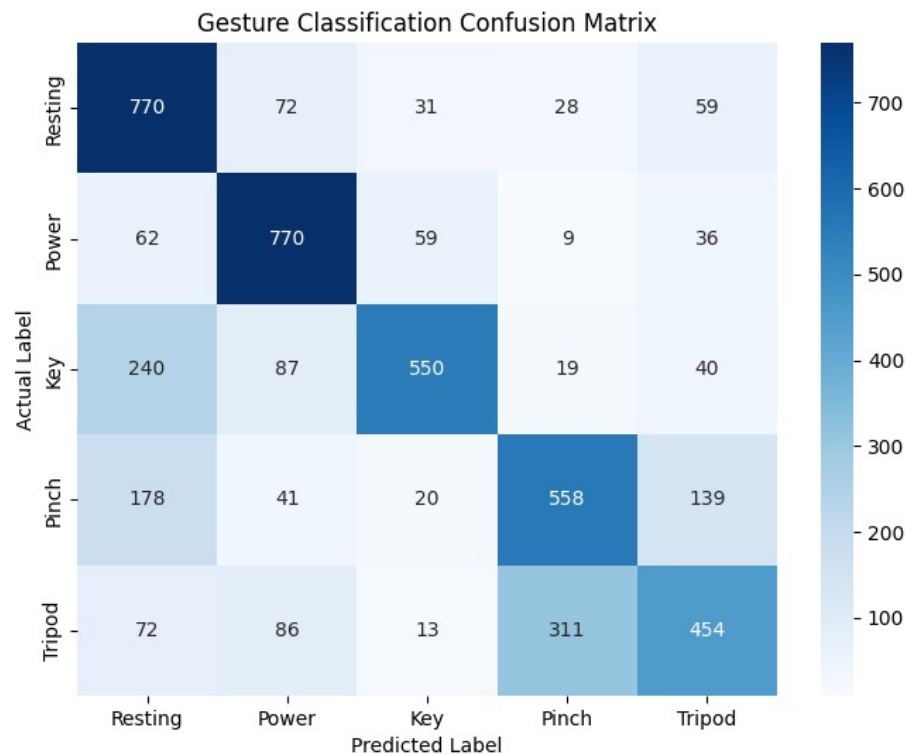
(venv) hayate:/var/home/hayate/Documents/Python$
```

Note:

- 2 Actual Predictions Made, Both Correct
 - (98% average accuracy over 20 tests)
- 8 Predictions Not Made
 - 4 Were Correct and Close To Confidence Threshold
 - 1 Was Correct and Not Close to Confidence Threshold
 - 3 Were Incorrect and Not Close to Confidence Threshold

Progress to Date

Created Generalized “Second Opinion” 1D Dilated CNN



Note:

- Only 67.7% accurate on classifications from more generalized datasets
- However, when used as a second opinion model and both models must vote on the most likely class, accuracy of the two model system boosts to an average of 85% on 20 tests.

Progress to Date



Put Entire Two Model System on Seeed ESP32-S3 Sense



Progress to Date



Put Entire Two Model System on Seeed ESP32-S3 Sense

```
=== Processing Window 6 ===  
  
>>> Final System Prediction: 4 <<<  
>>> Correct Prediction <<<  
  
=== Processing Window 7 ===  
  
>>> Final System Prediction: 0 <<<  
>>> Correct Prediction <<<
```

If the posterior probabilities of the classes exceed the confidence thresholds for the first model, it will act upon them quickly (without consulting the second model).

Progress to Date



Put Entire Two Model System on Seeed ESP32-S3 Sense

```
=== Processing Window 4 ===
Within 15% margin. Waking up Model 2 (CNN)...
Model 2 (CNN) Class Prediction
-> CNN agrees with RDA's initial hunch. Prediction ACCEPTED.

>>> Final System Prediction: 2 <<<
>>> Correct Prediction <<<

=== Processing Window 5 ===
Within 15% margin. Waking up Model 2 (CNN)...
Model 2 (CNN) Class Prediction
-> CNN disagrees (Predictions are far apart). Output remains REJECTED.

>>> Final System Prediction: -1 <<<
>>> Will Remain in Previous State <<<
```

If the posterior probabilities of the classes do not exceed the confidence thresholds for the first model but fall within a 15% margin below them, it will consult the second model.

Progress to Date

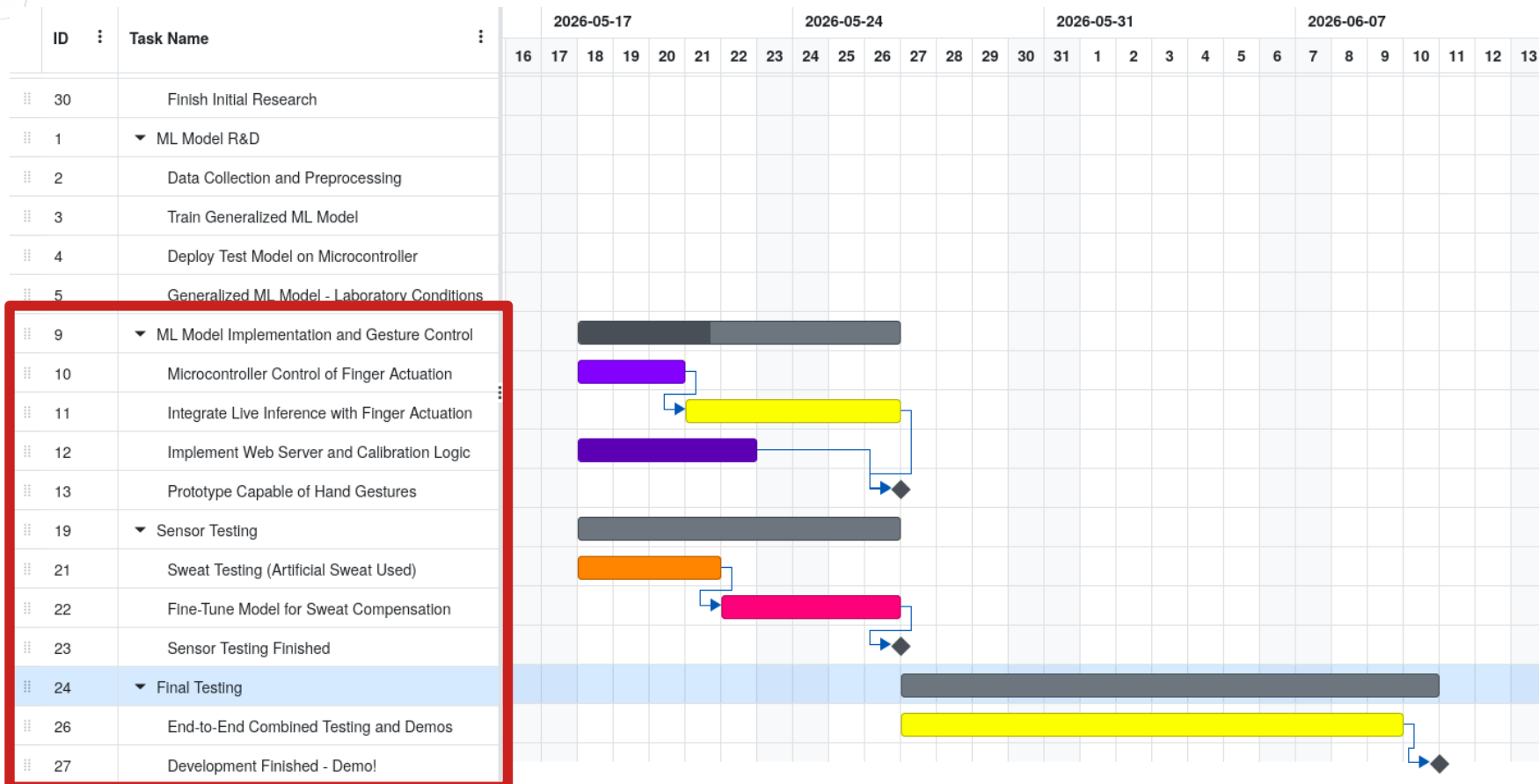


Put Entire Two Model System on Seeed ESP32-S3 Sense

```
=== Processing Window 1 ===  
  
>>> Final System Prediction: -1 <<<  
>>> Will Remain in Previous State <<<  
  
=== Processing Window 2 ===  
  
>>> Final System Prediction: -1 <<<  
>>> Will Remain in Previous State <<<  
  
=== Processing Window 3 ===  
Within 15% margin. Waking up Model 2 (CNN)...  
Model 2 (CNN) Class Prediction  
-> CNN disagrees (Predictions are far apart). Output remains REJECTED.  
  
>>> Final System Prediction: -1 <<<  
>>> Will Remain in Previous State <<<
```

If the posterior probabilities of the classes do not exceed the confidence thresholds for the first model by a wide margin (more than 15% for all) it will remain unchanged from its previous state (without consulting the second model).

Gantt Diagram



Next Steps



- 1) Collect more data (and refine and fine-tune generalized model)**
- 2) Make functions for finger actuation and connect ESP32-S3 Sense to prosthetic hand motors.**
- 3) Finish project this week and test and continue to refine model accuracy over the following weeks.**

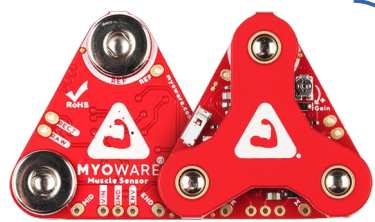
References



- 1) Jaramillo-Yáñez, Andrés, Marco E. Benalcázar, and Elisa Mena-Maldonado. 2020. "Real-Time Hand Gesture Recognition Using Surface Electromyography and Machine Learning: A Systematic Literature Review." *Sensors* 20(9):2467. doi:[10.3390/s20092467](https://doi.org/10.3390/s20092467).
- 2) Young, Peyton R., Kihun Hong, Eden J. Winslow, Giancarlo K. Sagastume, Marcus A. Battraw, Richard S. Whittle, and Jonathon S. Schofield. 2025. "The Effects of Limb Position and Grasped Load on Hand Gesture Classification Using Electromyography, Force Myography, and Their Combination." *PLOS ONE* 20(4):e0321319. doi:[10.1371/journal.pone.0321319](https://doi.org/10.1371/journal.pone.0321319).
- 3) Aarotale, Parshuram N., and Ajita Rattani. 2024. "Machine Learning-Based sEMG Signal Classification for Hand Gesture Recognition."doi: [10.48550/arXiv.2411.15655](https://doi.org/10.48550/arXiv.2411.15655)

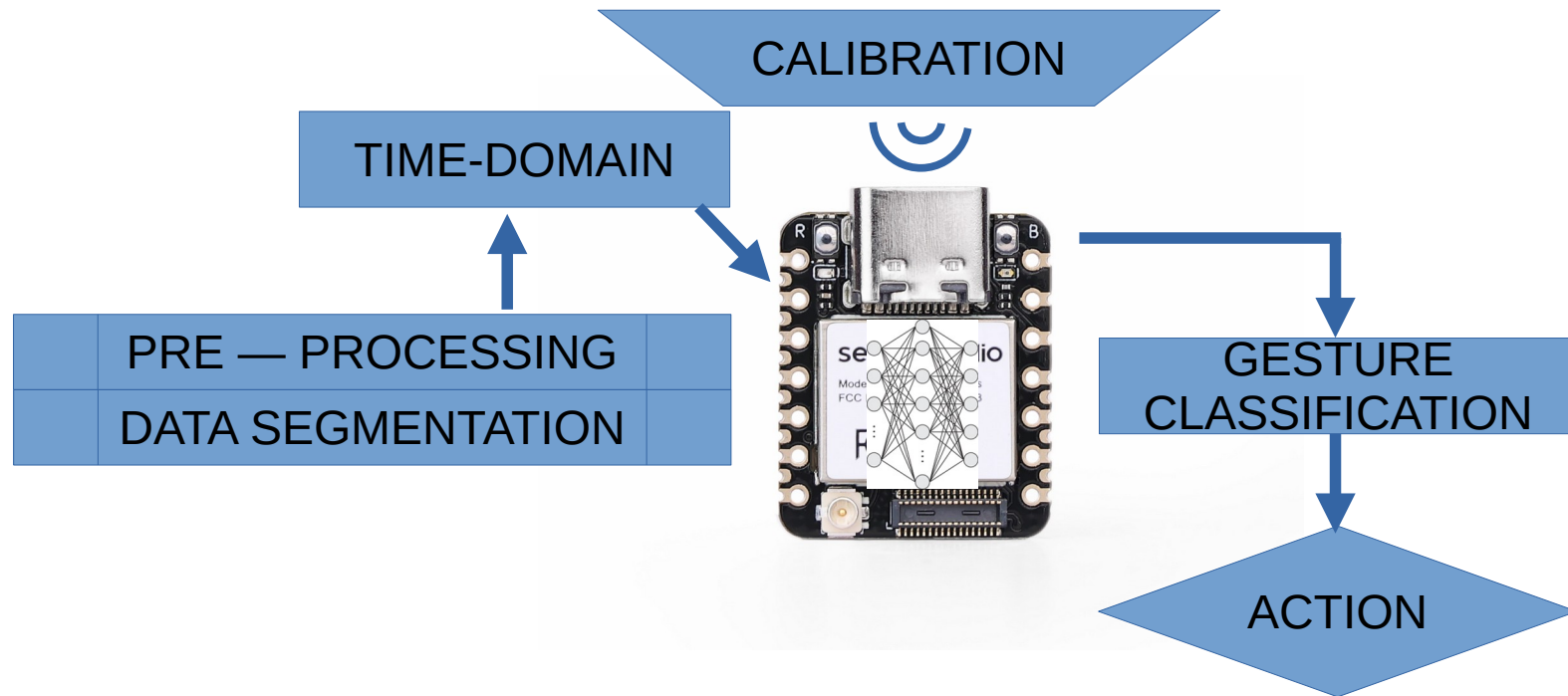
*All other images generated with Gemini AI or made by author.

Development Plan



EMG1

EMG2



TRAIN!